

Asset Management System — Tech Stack

Phase 1 | Free & Open-Source | March 2026

Stack

Layer	Tool	Why
Frontend	React 19 + TypeScript	React 19 Compiler reduces unnecessary re-renders by 30–60% · TypeScript catches bugs at compile time. Vite for fast builds · Free on Vercel
Mobile	Jetpack Compose	Build app for Android only
Backend	FastAPI + Pydantic	Same language as frontend · Shared types end-to-end · Free on Vercel.com
Database	PostgreSQL (Supabase)	Relational, ACID-compliant · Free 500MB hosted tier · Daily backups included
Auth & RBAC	Centralised Auth Service	Commonly built shared service · AMS consumes it, does not own it
QR Code	qrcode (npm)	Server-side generation · No external API · Outputs PNG · Free
Telemetry	Loguru, Uptime Kuma, Glitch Tip	Structured logs ·
Testing	Vitest + React Testing Lib + Supertest + Playwright	Unit, component, API & E2E coverage · All free · Runs in GitHub Actions CI
CI/CD	GitHub Actions + Docker	Free pipeline · Type-check + test + build + deploy on every push
Hosting	Vercel	Frontend on Vercel · Backend on Render · Both free tier

Known Limits in Phase 1

- Supabase free project pauses after 1 week of inactivity — needs a keep-alive ping
- Gmail SMTP capped at 500 emails/day — fine for Phase 1, swap to SendGrid in Stage 2
- Sentry free tier: 5,000 error events/month — sufficient for Phase 1 scale

Testing Strategy

Type	Tool	What It Covers
Unit	Vitest	Pure functions, utilities, business logic · Fast, Vite-native, TypeScript out of the box
Component	React Testing Library	UI components render correctly · User interactions work as expected
API	Supertest + Vitest	Every REST endpoint · Auth checks, role enforcement, error responses
E2E	Playwright codegen, pytest, vitest	Critical user flows end-to-end: login, assign asset, QR scan · Runs on real browser
Mobile	Jest	Android device

- All tests run automatically in GitHub Actions on every push · PR cannot merge if tests fail
- Coverage target: 80% on business logic · 100% on auth & role-enforcement paths

Stage 2 Upgrades

Layer	Stage 1	Stage 2 Upgrade
Mobile	Android app (jetpack)	Share app other
Logs	Loguru, Uptime Kuma, Glitch Tip	Logtail or Datadog — searchable log UI
Telemetry	Sentry + UptimeRobot	OpenTelemetry + Grafana dashboard